Decision-Level Road Network Fusion From Multisource Remote Sensing Data via Evidential Reasoning Rule and Topology Constraints

Fanghong Xiao, *Member, IEEE*, Ling Tong, *Senior Member, IEEE*, Shiyu Luo, *Member, IEEE*, Yuchuan Wang, and Jiaxing Yang

***Abstract*—Recent decades have seen significant advances in road network extraction via artificial intelligence. However, single-source approaches remain vulnerable to sensor-specific limitations, such as occlusions in optical imagery. While multisource fusion can mitigate these gaps, sensor heterogeneity and the lack of well-registered, well-labeled multisource datasets make establishing a unified fusion strategy difficult. In this article, we propose a generic framework for fusing road information derived from multisource remote sensing (RS) images at the decision level, enabling effective combination of outputs from diverse single-source machine learning (ML) models. The presented method consists of two stages. First, to address redundancy (RD) from repeated extractions, we propose a geometric rule-based fusion strategy, featuring a novel nonmaximum suppression (NMS) algorithm tailored for linear targets. Second, evidence from different sources is aggregated using the evidential reasoning (ER) rule, guided by each ML model's performance and the confidence maps of their outputs. The final fusion result is achieved by optimizing graph labeling through an energy function that integrates the ER rule with topological constraints (TCs). The proposed framework effectively accommodates heterogeneous inputs and is validated on multitemporal and multimodal RS images. Experimental results demonstrate marked improvements in road network completeness (CP) and correctness (CR) over single-source extraction, while also outperforming representative decision-level fusion methods in overall quality (QL) with reduced RD.**

***Index Terms*—Decision-level fusion, evidential reasoning (ER) rule, remote sensing (RS), road network extraction, topology.**

NOMENCLATURE

B^L	Set of all line segments (edges).
B^L_{gf}	Set of edges after geometry-driven fusion.
d_θ	Included angle between line segments.
f_r	Overlap ratio between line segments.
e_i	ith line segment.
$\|e_i\|$	Pixel length of e_i.
ζ_i	Normalized pixel length of e_i.
θ_i	Set of angles between e_i, and its neighbors.
$C(e)$	Mean confidence score of pixels in e.
$p_{h,j}$	Belief probability of class h under source j.
r_j	Reliability of source j.
w_j	Weight of source j.
$p_{h,E(2)}$	Combined belief probability for class h.
l_1	Label of e_i.
U, U_d, U_p	Total energy function; observation energy function; priori energy function.
γ	Weighting coefficient of energy function.
u_p	Local priori energy function.
Θ_i	Minimum normalized angle between e_i and its neighbors.
$d(v)$	Degree of vertex v.

I. INTRODUCTION

A ROAD is a critical infrastructure essential for all types of vehicles. Accurate acquisition of road geospatial information is of great importance for urban planning, land utilization, vehicle navigation, disaster emergency management. and other applications [1]. Traditionally, road location information is collected using the global positioning system (GPS) receiver installed on the ground mobile platform [2]. These locations are linked together to form a road network. The ground-based method for surveying and monitoring road networks usually has a high positioning accuracy, but it is laborious and time-consuming. Thanks to the substantial progress in remote sensing (RS) technology and artificial intelligence in recent decades, automatic road network extraction from airborne or spaceborne RS images has become feasible [3], [4]. Satellite imagery offers wide coverage, ranging from several to hundreds of kilometers, and RS sensors can provide near real-time data for dynamic monitoring.

In view of these advantages, road extraction with RS images, including road network extraction, has attracted significant attention. As of 2025, approximately 10 000 papers on road extraction from RS images can be retrieved in the Web of Science database using the query "TS = (RS AND

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road).” Notably, about 90% of the state-of-the-art literature focus on single-source RS data. However, methods based solely on single-source RS data often encounter insurmountable bottlenecks. For instance, optical images are susceptible to weather conditions and shadows, while synthetic aperture radar (SAR) data suffers from speckle noise and geometric distortions. These “information blind spots” make it difficult for any single sensor to provide robust detection in complex environments. Therefore, multisource data fusion is essential to exploit complementary information and improve robustness.

Despite its clear advantages, the development of multisource fusion methods for road extraction remains limited. This is largely due to the limited availability and high cost of coregistered datasets, as well as the complexity of joint annotation. In addition, the intensive computational demands associated with large-scale processing may also impose significant constraints. Most critically, robust and generalizable fusion of multisource RS data remains a major technical challenge, underscoring the need for more effective methodologies.

Although multisource fusion for road extraction remains limited and faces significant challenges, multisource data fusion in the broader RS community has been continuously studied for over two decades [5], [6], [7]. Beyond road extraction, data fusion applications also encompass land-use classification, biomass estimation, 3-D reconstruction, and other domains [8]. Generally, data fusion can be divided into three types: raw data level, feature level, and decision level. Among these, decision-level fusion attracts particular attention due to its flexibility in integrating heterogeneous data sources without requiring strict spatial coregistration or uniform feature representations [9]. In this context, various ensemble strategies have been presented. In [10], multiple extraction results are directly superimposed, which offers simplicity but introduces considerable redundancy (RD) from repeated extractions. Similar static rule-based fusion methods, such as majority voting [11] and Borda count [12], are also easy to implement but exhibit limited performance, as they disregard prior knowledge and the predictive reliability of individual sources. To remedy these drawbacks, evidence-based decision-level fusion methods have been introduced. A representative approach is Bayesian decision theory [13], [14], which provides a unified mathematical framework but depends critically on accurate probability estimation. Machine learning (ML) methods, e.g., support vector machines (SVMs) [15], [16], random forest [17], and deep neural networks (DNNs) [18], [19], have also proven effective in multisource decision-level fusion tasks. They are capable of modeling complex nonlinear relationships, yet their performance relies on sufficient well-registered training datasets. In addition, Poulain et al. [20], Belmahdi et al. [21], and Chen et al. [22] proposed solutions based on Dempster–Shafer (DS) evidence theory, which is also a widely used data fusion approach [23]. It offers the advantage of explicitly representing uncertainty. However, it is sensitive to highly conflicting evidence and treats all information sources equally by default.

From the perspective of data source types, commonly used multisource data in RS research primarily include multitemporal and multimodal images. For multitemporal data, images of the same scene acquired on different dates can be fused to improve detection accuracy. Regarding road network extraction, an early study employed fuzzy fusion techniques for linear features using multitemporal SAR images [24]. In [25], a classification strategy based on the histogram distance index and a neural classification tool is presented to exploit the temporal RD of a sequence of satellite SAR data. Benefiting from the coherence characteristics of multitemporal interferometric SAR data, Jiang et al. [26] and Xiao and Tong [27] proposed the combination of intensity and coherence features to obtain superior road network extraction results, respectively. More recently, deep learning models [28], [29] were introduced for road mapping via multitemporal SAR images, leveraging its strong learning ability for complex features.

Multimodal RS image fusion has also been explored, as combining multiple modalities allows for more detailed and precise scene characterization that cannot be achieved with single-modality data [30]. For example, through formulating prior knowledge about the scene objects with semantic nets, Tönjes and Growe [31] presented a knowledge-based road network extraction approach by combining cues from aerial optical, infrared, SAR images. As discussed in [32] and [33], evidence theory is applied to fuse SAR measurements and optical images in order to update the road network data in geographic information systems. Other early approaches for road network extraction using multimodal data include fusion based on spectral characteristics and priori rules [34], region growing with road seeds obtained from different sources [35], and fusion using the general form of symmetrical sum to combine line detection responses [36]. In recent years, cross-modality learning has gained traction. Lin et al. [37] developed a UU-Net architecture that fuses deep features from Sentinel-1 (SAR) and Sentinel-2 (optical) imagery, achieving improved accuracy in large-scale road extraction. Filho et al. [38] introduced an early-fusion multimodal model that jointly processes RGB images and digital surface models (DSMs) to enrich both contextual and geometric road cues. More recently, hybrid approaches incorporating high-resolution RS imagery with GPS trajectory maps have been explored [39], [40], [41].

Overall, most of the above studies for road extraction, particularly those based on deep learning, predominantly employ feature-level fusion methods. Beyond the challenges of aligning heterogeneous features, feature-level fusion often results in a tightly coupled architecture that lacks flexibility and generalizability across diverse geographic regions or sensor configurations. Moreover, deep-learning-based fusion approaches typically demand large-scale, coregistered, and well-annotated multimodal training datasets. It is both costly and inflexible, as different combinations of multisource inputs often require training separate models. In contrast, decision-level fusion offers greater modularity and robustness by aggregating independent predictions from individual modalities. However, existing decision-level fusion methods also face limitations, such as neglecting contextual or topological prior information, disregarding the reliability of information sources, and failing to account for modality-specific uncertainties. These oversights can significantly degrade fusion performance, especially in complex scenarios.

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Motivated by these challenges, we propose a general decision-level fusion framework for road network extraction, integrating multisource predictions with explicit modeling of road structural priors. Unlike conventional methods that separately perform evidence fusion and topological refinement, the proposed approach jointly models belief estimation, multisource fusion, and topological constraints (TCs) within a unified framework. It further mitigates bias in learning-based belief assignments and enables topology-aware fusion on structured road segments. Importantly, the framework remains effective even in the absence of coregistered multisource training data. The main contributions are highlighted as follows.

1) We propose a generic two-stage decision-level fusion framework for the road network extraction from multisource RS images. In this framework, the outputs of ML models applied to diverse source images are effectively fused to generate a more complete road network map. Experimental results with multitemporal and multimodal images demonstrate that the proposed framework achieves superior accuracy and exhibits strong potential for practical engineering applications.
2) A novel geometric-based fusion scheme, namely line-specific nonmaximum suppression (Line-NMS), is proposed to address the problem of overlap and RD in the road extraction results of multisource RS images. Specifically, road segments from different sources are merged into nonredundant segments based on their spatial overlap and geometric proximity.
3) We propose a topology-aware road fusion approach based on the evidential reasoning (ER) rule. For each RS source, belief probabilities of road segments are estimated via a sigmoid-based fitting strategy learned from confidence score histograms to reduce potential bias. The ER rule is then applied to fuse multisource evidence, while TCs are incorporated into a graph optimization formulation to improve road connectivity.

The rest of this article is organized as follows. In Section II, the related work is introduced. The proposed method with a general framework is presented in detail in Section III. The case studies, comparison analysis, and discussion are shown in Section IV. The conclusions are given in Section V.

II. Related Work

A. ER Rule

The ER rule extends the classical DS combination rule to mitigate its well-known sensitivity to highly conflicting evidence. In the field of RS, DS theory has been applied to tasks such as change detection [42] and target recognition [43]. Although DS theory represents uncertainty via belief assignments over a frame of discernment, it implicitly assumes all sources are equally reliable and normalizes away conflict. This often leads to counterintuitive fusion results when strong disagreement exists among sources. In contrast, the ER rule explicitly incorporates source reliability and weight, and redistributes conflict according to source credibility, leading to more robust fusion under uncertainty [44].

The ER algorithm has been widely applied in system analysis and decision-making problems [45]. For example, Ning et al. [46] proposed an ER-based health assessment method that effectively fuses nonequidistant subsystem data using a discount coefficient to preserve critical information and enhance accuracy. Inspired by the effectiveness of ER theory in reducing uncertainty during information fusion, Qin and Zeng [47] proposed a product ranking method based on ER rule to aggregate multiclassifier sentiment analysis results from online reviews. Liu et al. [48] introduced a fault detection algorithm based on ER that jointly considers information reliability and uncertainty by introducing a novel basic belief assignment method in the complex domain. These studies demonstrate the advantages of ER over the classical DS combination rule in multisource information fusion.

In addition to source reliability and weight, the initial belief probability (BP) also plays a key role in determining the final fusion result. However, existing studies often overlook the potential bias in the original BP, particularly when soft classification probabilities from ML models are directly adopted as belief assignments. To address this issue in road fusion tasks, our method learns sigmoid-shaped BP functions from the histograms of confidence scores on training samples for each source. It is a purely data-driven approach without making any assumptions about the relationships between different sources and classes.

B. Road Topology

Topological relationships describe the qualitative properties that characterize the relative position of spatial objects. In real-world scenarios, road systems exhibit distinct and well-defined topological characteristics. Typically, roads are not isolated without functional reasons; instead, they are designed to connect and form a network. The end of a road is either connected to another road segment, or connected to a built-up area. Besides, the connections between road segments usually appear as the shape of "+," "T," "L," or "Y" [49]. Although objects such as vehicles, overpasses, buildings, and trees may occlude roads in images, leading to unclear boundaries or local discontinuities, road networks still exhibit strong global connectivity and structural regularity.

To exploit this prior knowledge, researchers have attempted to reframe road network construction as a graph problem [10], [36]. There are various methods to represent a road network using a graph, where the nodes can correspond to road junctions, segments, continuous angle roads, routes, and so on [50]. For example, Li et al. [51] introduced a structure-aware road extraction framework that explicitly refines road connectivity and corrects network topology, enabling more accurate and topologically consistent road network reconstruction. Wu et al. [52] proposed a CPP-based Bayesian road extraction method that explicitly models and constrains road network topology by feature point processes. Wei et al. [53] proposed a multitask learning framework that jointly performs road detection and centerline extraction, introducing a road-topology-aware loss to explicitly enforce connectivity and suppress gaps and spurious segments. Saati and Amini [54] presented an SAR-based road extraction method that constructs road networks by

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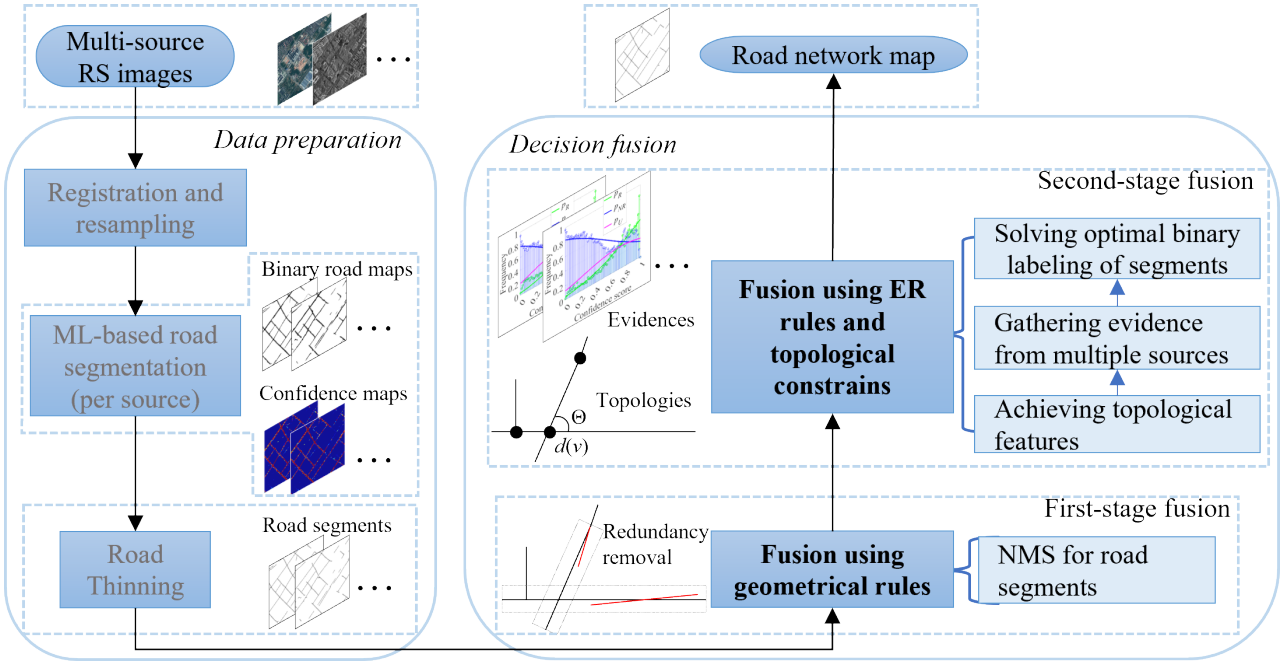


Fig. 1. Overview of the proposed decision fusion framework for road network extraction.

connecting thinned road segments via a network snake model with energy minimization and perceptual grouping.

Despite these efforts to incorporate topological information, most existing methods focus on single-source data and introduce TCs as a postprocessing step to refine initially extracted road segments. In contrast, our approach embeds road topology constraints directly into the multisource fusion framework by designing an energy function that explicitly encodes topological relationships. This ensures that topological correctness (CR) is enforced during fusion, rather than after reconstruction.

III. Methods

The proposed method deals with multisource RS images, including standard airborne or spaceborne RS image products. These products can be data observed from different dates or sensors. Fig. 1. gives an overview of the decision-level fusion framework for road network extraction. The framework consists of two steps. The first is data preparation, which is further specified into image registration, road segmentation using AI models, and road thinning to achieve skeleton lines, as discussed in Section III-A. The second is a two-stage decision-level fusion, which includes fusion based on geometrical rules and fusion combining ER rule and topological constrains. The latter is the focus of this article and is described in detail from Sections III-B to III-E. The steps not specifically elaborated on in this article are highlighted in light gray in Fig. 1. In addition, the main notations used throughout this article are summarized in Nomenclature.

A. Data Preparation

The objective of this article is decision-level fusion of roads, which requires image registration and resampling as a prerequisite. The registration step ensures that the images are spatially comparable and geometrically consistent. Here, the image registration is not specifically discussed. We previously registered the multisource images by using affine transformation [55] with manually selected control points. Notably, compared to data- or feature-level integration, our decision-level fusion framework is inherently more robust to minor spatial misalignments as it operates on object-level road segments.

In view of the fact that a large number of ML-based road segmentation models for single-source RS images have been proposed in the past few years [56], the idea underlying this article is to improve the quality (QL) and reliability of road extraction by fusing various segmentation results. Hence, for registered multisource RS images, it is important to select a road segmentation model that aligns with the characteristics of each image source. In this study, DLinkNet34 [57] and MSMDFF-Net [58] are adopted as the road segmentation methods for the decision-level fusion experiments. The selection of these two distinct architectures is intended to evaluate the versatility and generalizability of the proposed fusion framework across different segmentation paradigms. The resulting segmented road maps, along with the corresponding confidence maps generated by the ML models, serve as inputs for the subsequent steps.

The road thinning [59] process is to achieve the line segments of roads, which serve as the basic operation units for the proposed fusion method. Previous studies have shown that the vectorial road segments can reduce the computational load and contribute to the construction of road network topology [49]. In our previous work, we proposed a road centerline extraction method from the segmented road map based on perceptual grouping and curve fitting [60]. This method is adopted here because it shows merits in extracting accurate

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Algorithm 1 Line-NMS

Input: Initial line segment set B^L
Output: Fused line segment set B^L_{gf}
S1: Initialize $B^L_{gf} = \varnothing$
S2: Select the longest segment $e_{\max}$ from B^L;
  Update $B^L = B^L - \{e_{\max}\}$;
  Update $B^L_{gf} = B^L_{gf} \cup \{e_{\max}\}$
S3: Initialize a temporary set $B^L_{temp} = B^L$;
  for each segment e in B^L **do**
    Compute $\theta = d_\theta(e, e_{\max})$ using (1)
    if $\theta < \theta_{th}$ **then**
      Update $B^L_{temp} = B^L_{temp} - \{e\}$;
      Compute $r = f_r(e, e_{\max})$ using (2);
      Compute $r_{th}(e)$ using (3)
      if $r \le r_{th}(e)$ **then**
        Update $e = e - \left(e \cap e^{ex}_{\max}\right)$;
        Update $B^L_{temp} = B^L_{temp} \cup \{e\}$
      end if
    end if
  end for
  Update $B^L = B^L_{temp}$
S4: Repeat S2 and S3 until $B^L = \varnothing$

thinning line segments and their corresponding road width information. More importantly, it demonstrates adaptability for road segmentation maps from different sources.

B. Fusion With Geometrical Rules

Since the road segmentation map obtained from different source images corresponds to same geographic area, it is foreseeable that there may exist a large amount of RD in the merging of road segmentation results. The geometric-driven fusion aims to reserve different parts and solve duplicate parts. More specifically, road segments from different sources are combined into nonredundant line segments if they are overlapped and close enough.

Let B^L_i be the set of line segments obtained from the RS image of source i. Assume $B^L = B^L_1 \cup B^L_2 \cup \cdots \cup B^L_N$ is the set which contains all the line segments from all the N sources. Our geometry-driven fusion method, termed Line-NMS, is carried out as described in Algorithm 1. It is inspired by the principles of nonmaximum suppression (NMS) [61]. We first initialize B^L_{gf} as an empty set, which is used to iteratively store line segments after the fusion operation. In order to eliminate redundant line segments, we define the included angle d_θ and overlap ratio f_r between line segments. Given line segments e_1 and e_2, d_θ is computed as follows:

$$d_\theta(e_1, e_2) = \min\left\{\arccos\left\langle\overrightarrow{o_1}, \overrightarrow{o_2}\right\rangle, \pi - \arccos\left\langle\overrightarrow{o_1}, \overrightarrow{o_2}\right\rangle\right\} \quad (1)$$

where $\overrightarrow{o_1}$ and $\overrightarrow{o_2}$ are the vectors determined by the two endpoints of e_1 and e_2, respectively. The overlap ratio of e_1 with respect to e_2 is defined as follows:

$$f_r(e_1, e_2) = card\left(e_1 \cap e_2^{ex}\right) \Big/ card(e_1) \quad (2)$$

where *card* is the number calculation of pixel elements in a set, $\cap$ denotes the intersection operator, and e_2^{ex} is the dilation result of e_2 according to its width feature achieved by the RoC detector in [60]. Fig. 2 presents the sketch map related to the included angle d_θ and overlap ratio f_r.

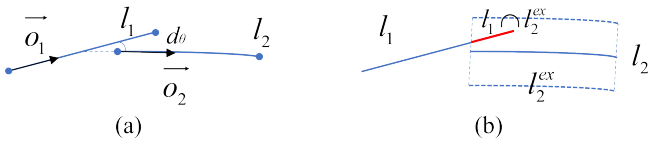


Fig. 2. Schematic illustrations of the definitions used in Line-NMS. (a) Included angle d_θ. (b) Overlap ratio f_r.

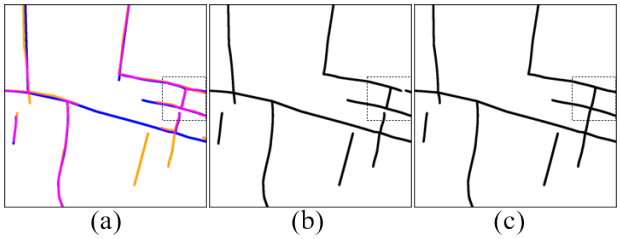


Fig. 3. Road fusion using geometrical rules (dotted boxes highlight areas for comparison). (a) Road segments obtained from two different sources (orange and purple parts are from source 1, blue and purple parts are from source 2). (b) Result after applying Line-NMS. (c) Result after applying junction regularization.

In Algorithm 1, the parameter θ_{th} is aimed at restricting the orientation search range when identifying redundant segments in B^L, and is fixed to $\pi/8$ in this article. The second parameter r_{th}, which is used to determine RD, is computed adaptively by

$$r_{th}(e) = |e|/|e_{\max}| \quad (3)$$

where $|e|$ is the pixel length of e. As the length of e decreases, the corresponding threshold r_{th} also reduces. Taking the road segment maps from two sources in Fig. 3(a) as an example, Fig. 3(b) shows the results of Line-NMS where overlapped road segments are merged.

To correct the road network topology errors caused by the process of road thinning and Line-NMS, a junction regularization is further applied. Specifically, a method based on spline fitting [60] is applied on B^L. Endpoints and intersections of line segments that lie within a close proximity are merged into a single junction. This ensures better connectivity between line segments and restores accurate junction structures. An example of the regularization outcome is shown in Fig. 3(c), corresponding to the input in Fig. 3(b).

C. Achieving Topological Features From the Road Network

In order to incorporate topological constraints into the road fusion model, we construct a graph representation of the road network. The graph consists of vertices connected by edges loaded with attributes and labels. The graph construction approach is quite basic and relies on connected-component analysis [62] and Douglas–Peuker (DP) algorithm [63]. The overall process primarily involves the following.

1) Detecting all road intersections and segmenting at each intersection.

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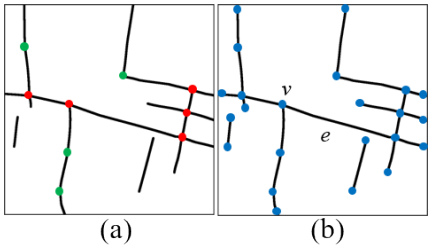


Fig. 4. Graph construction. (a) Intersections (red dots) and turning points (green dots). (b) Final graph map composed of vertices (blue dots) and edges (black lines).

2) Labeling the connected component on the road map.
3) For each component, applying DP algorithm to detect the internal turning point and segmenting at the turning point.
4) Finally, constructing the graph where connected components are treated as edges (e) and all intersections, endpoints, and turning points become vertices (v).

The DP algorithm, originally designed for curve simplification, is used here to identify significant turning points. This process ensures that each resulting segment exhibits low curvature. Fig. 4 shows an example of the proposed graph construction.

Based on the constructed graph, we then extract topological features. Specifically, the degree of each vertex v, denoted as $d(v)$, is defined as the number of edges associated with v. For each edge e_i, normalized length is calculated by the following formula:

$$\zeta_i = \min\{|e_i|/D_{\max}, 1\} \tag{4}$$

where $|e_i|$ is the pixel length of e_i, and D_{max} is the normalized length parameter. In addition, angle attributes are computed for e_i using

$$\theta_i = \{\theta | \theta = \angle e_i e, \forall e \leftrightarrow e_i\} \tag{5}$$

where $e \leftrightarrow e_i$ represents that two edges share a vertex. By assigning labels to the edges, road network extraction is reformulated as the task of solving optimal road labels to the edges within a graph representation.

D. Acquiring Evidences From the Multisource Data

1) Assumptions: This article aims to develop an ER-based fusion method for the road network extraction problem. For multisource data, our method relies on the assumptions listed below.

1) Each source outputs a road candidate map and the map can be further processed as a road map composed of thinned line segments.
2) Each source can also output a confidence score map corresponding to the road candidate map.

Assumption 1) indicates that the fusion occurs at both the decision level and the object level. Assumption 2) identifies the sources of evidence utilized in the ER rule-based fusion process.

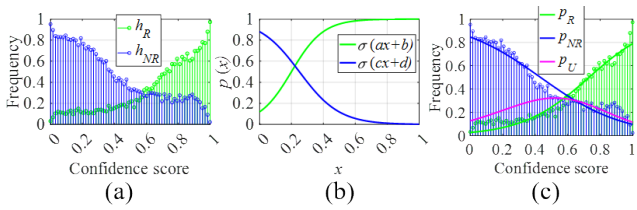


Fig. 5. Definition and determination of BP functions. (a) Examples of histograms for confidence scores. (b) Model for BP functions. (c) BP functions with optimized parameters for Fig. 5(a).

2) BP Function Definition: Suppose that $H = \{R, NR, U\}$, where R represents the class of road, NR is the nonroad class, and U denotes uncertainty. According to the theory of ER rule [47], it is essential to associate each class of each source with a BP. Unlike existing methods that often directly adopt biased soft classification probabilities, we propose a data-driven estimation phase. Specifically, sigmoid-shaped BP functions are learned from the confidence score histograms of training samples for each source. This calibration process handles modality-specific biases without prior assumptions, ensuring that BP assignments are consistent and comparable across heterogeneous models.

Let us assume that C is a confidence map deduced from an ML model. Let $(C(e), y)$ be the sample set, where $y \in H, e$ represents line segment, $C(e)$ denotes the mean confidence score of pixels in e. In order to obtain continuous BP function from discrete sample points, curve fitting is adopted using histograms of $C(e)$. More precisely, for a suitably chosen number M, we first define bin boundaries $0 = x_1 \leq x_2 \leq \cdots x_m \cdots \leq x_{M+1} = 1$, where x_m represents confidence scores and the bin X_m is defined by the interval $(x_m, x_{m+1}]$. Then, the frequency of each class (h_R, h_{NR}, h_U) in each interval can be statistically obtained by

$$\begin{cases} h_R(X_m) = n_m^R / n_m \\ h_{NR}(X_m) = n_m^{NR} / n_m \\ h_U(X_m) = 1 - h_R(X_m) - h_{NR}(X_m) \end{cases} \tag{6}$$

where n_m denotes the number of all samples for which $C(e) \in X_m$, and n_m^R and n_m^{NR} are the number of road and NR samples for which $C(e) \in X_m$, respectively. As a result, the histogram shown in Fig. 5(a) can be achieved.

Different from most methods that define the BP function as a trapezoidal function [64], this article proposes to define the BP function based on the sigmoid function [Fig. 5(b)] as follows:

$$\begin{cases} p_{R,j}(x) = \sigma(a_j x + b_j) \\ p_{NR,j}(x) = \sigma(c_j x + d_j) \\ p_{U,j}(x) = 1 - p_{R,j} - p_{NR,j} \end{cases} \tag{7}$$

where $\sigma(x) = 1/(1 + e^{-x}), a_j > 0, c_j < 0$, $p_{R,j}$ represents the BP value of class R corresponding to the confidence score x under the evidence source j, and $p_{NR,j}$ and $p_{U,j}$ are the same. One of the reasons why we choose the sigmoid function is that it has higher fitting accuracy and is more suitable for the real histogram distribution compared with the trapezoidal function.

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On the other hand, the sigmoid function is simple and easy to solve the optimal parameters.

The BP functions rely on four parameters $\varphi_j = (a_j, b_j, c_j, d_j)$ [as presented in (7)], where j corresponds to each source. These parameters can be obtained empirically or optimized based on ground truth data. It is worth noting that, once optimized with a specific dataset, the BP functions can be applied to other images acquired from the same source.

3) Parameter Optimization of the BP Functions: For each source, three BP functions are defined to represent the classes of road, NR, and uncertainty, resulting in four parameters to be optimized per source. As described in the previous section, parameters optimization can be conducted in the case of an available sample set. Using (6), the point set $\{(\overline{x}_m, h_R(X_m))\}, \{(\overline{x}_m, h_{NR}(X_m))\}$ can be obtained, where $\overline{x}_m = (x_m + x_{m+1})/2, m = 1, 2, \ldots, M$. We use those point sets to optimize the parameters of the BP functions, which is subsequently applied to perform the fusion of roads extracted from different sources. The objective function for estimating φ_j is defined as follows:

$$\begin{aligned} \min_{\varphi_j} \; & \left\{ \sum_{m=1}^{M} [p_{R,j}(\overline{x}_m) - h_{R,j}(X_m)]^2 \right. \\ & \left. + \sum_{m=1}^{M} [p_{NR,j}(\overline{x}_m) - h_{NR,j}(X_m)]^2 \right\} \\ \text{s.t. } & a_j > 0, c_j < 0 \\ & p_{R,j}(x) + p_{NR,j}(x) \le 1 \quad \forall x \in [0, 1]. \end{aligned} \tag{8}$$

This is a nonlinear least squares problem with nonlinear constraints, and thus, can be solved based on the Levenberg–Marquardt algorithm [65]. For the case of Fig. 5(a), this procedure leads to the optimized BP function in Fig. 5(c).

4) Aggregation Based on the ER Rule: According to the ER rule, each piece of evidence E_j is characterized by a reliability r_j and a weight w_j. r_j $(0 \le r_j \le 1)$ is an inherent property of evidence, which measures the ability of the information source to provide correct result. w_j $(0 \le w_j \le 1)$ represents the importance of evidence for users. Based on these definitions, the weighted BP with reliability is defined by

$$\hat{m}_{h,j} = \begin{cases} 0, & h = \emptyset \\ c_{rw,j} m_{h,j}, & h \subseteq H, h \ne \emptyset \\ c_{rw,j}(1 - r_j), & h = P(H) \end{cases} \tag{9}$$

where $H = \{R, NR, U\}, m_{h,j} = w_j p_{h,j}, c_{rw,j} = 1/(1 + w_j - r_j)$, and $\hat{m}_{P(H),j}$ is the residual BP uncommitted by E_j.

For two evidences E_1 and E_2, the ER fusion rule is applied to achieve the combined belief probability $p_{h,E(2)}$, shown as follows:

$$p_{h,E(2)} = \begin{cases} 0, & h = \emptyset \\ \dfrac{\hat{m}_{h,E(2)}}{\sum_{A \subseteq H} \hat{m}_{A,E(2)}}, & h \subseteq H, \; h \ne \emptyset \end{cases} \tag{10}$$

where $\hat{m}_{h,E(2)} = [(1 - r_2)m_{h,1} + (1 - r_1)m_{h,2}] + \sum_{B \cap C = h} m_{B,1} m_{C,2}$.

For more than two pieces of evidence, the ER rule can be applied recursively to combine multiple sources, as detailed in [44].

E. Fusing With ER Rule and TCs

Thus far, according to the ER aggregation, the road candidate is most likely assigned to the class with the highest BP. However, the initial ER-based fusion cannot properly model the topological relationship between road segments. As illustrated in Section II-B, topological property is one of the key features of road besides spectral and texture features. Therefore, incorporating TCs into the fusion process is essential for improving road network extraction. Here, we introduce an improved fusion model based on the graph representation of road candidates.

Let $l_i \in \{R, NR\}$ be the hypothesis label of the ith $(i = 1, 2, \ldots, N)$ segment where R denotes "road" and NR is "nonroad." Suppose the label configuration $L_E = \{l_1, l_2, \ldots, l_N\}$ be the set of labels of all edges in graph G built from Section III-C. Obviously, L_E has 2^N different label configurations. Therefore, road network extraction can be transformed into the process of obtaining the optimal label L_E by minimizing the energy function. Accordingly, the following energy function is introduced:

$$U(L_E) = (1 - \gamma) \cdot U_d(L_E, D_E) + \gamma \cdot U_p(L_E) \tag{11}$$

where D_E is the set of mean confidence scores corresponding to all line segments, U_d is the observation energy function, U_p is the a priori energy function, and γ is a weighting coefficient used to balance the two terms. Based on sensitivity analysis, γ is typically set within the range of $[0.3, 0.5]$ to ensure a stable trade-off between observational evidence and structural priors.

The first term U_d measures the possibility of the segment that belongs to a road using evidences from multisource data and is given by

$$U_d(L_E, D_E) = \sum_{i=1}^{N} -p_{l_i, E(M)} \tag{12}$$

where $p_{l_i,E(M)}$ is the combined BP that the ith line segment belongs to label l_i, obtained based on (10) using confidence scores from M sources.

The second term U_p is related to the TCs of road segments. In order to penalize the label configuration that does not conform to the road topology characteristics (i.e., short-isolated roads, high vertex degree, and large curvature), U_p is calculated as the sum of the local energy u_p

$$U_p(L_E) = \sum_{i=1}^{N} u_p(l_i). \tag{13}$$

When the configuration of segment i is NR, u_p is defined as follows:

$$u_p(l_i = NR) = \begin{cases} -P_b, & \min_{v \in V_i} d(v) = 1 \text{ and } \max_{v \in V_i} d(v) > 1 \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where V_i denotes the set of vertices associated with i, and $d(v)$ denotes the degree of vertex v. Inversely, if segment i is labeled as R, u_p is expressed as follows:

$$u_p(l_i = R) = \begin{cases} -\zeta_i + P_i, & \max_{v \in V_i} d(v) = 1 \\ -\zeta_i \Theta_i^{\alpha}, & 1 < \max_{v \in V_i} d(v) \le 4 \\ -\zeta_i \Theta_i^{\alpha} + P_s, & \max_{v \in V_i} d(v) > 4. \end{cases} \tag{15}$$

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In (15), ζ_i denotes the normalized pixel length of the segment, which can be derived by (4), and $\Theta_i = \min\{\theta_i/\pi\}$ represents the minimum normalized included angle between segment i and its adjacent segments. The parameters P_b, P_i, P_s, and α are positive and serve as topological penalty coefficients designed to regularize the graph structure based on the intrinsic geometric properties of road networks.

1) P_b: It constrains the formation of short dangling edges (burrs) that often arise from noise in single-source segmentation. Physically, it ensures the continuity of the main road skeleton.
2) P_i: It is assigned a relatively higher value to suppress short, disconnected segments that lack topological support. This reflects the physical characteristic that roads are inherently connected infrastructures rather than fragmented clusters.
3) P_s: It penalizes intersections with unreasonable geometries. By assigning costs to unlikely node degrees or junction types, it encourages the fusion result to adhere to standard road layout patterns.
4) α: This parameter regularizes the directional smoothness of road segments. It ensures that the fused network maintains realistic curvature and avoids sharp, nonphysical turns at junctions.

In this article, the four aforementioned parameters are set a priori based on the characteristics of road network topology: $P_b = 0.3$, $P_i = 1.2$, $P_s = 0.3$, and $\alpha = 0.2$. These values have been found to be generally applicable to a wide range of urban road network extraction tasks. They can be further fine-tuned according to the specific spatial patterns or structural complexities of road networks in different application scenarios to achieve optimal performance. More details of these parameters can be found in Section IV-F.

Using the above definition of the energy function, the solving process of the optimal label configuration is realized by simulated annealing (SA) method [66]. In this article, the SA process is initialized with all labels assigned as road (R). The initial temperature, cooling rate, and terminal temperature are set to 1, 0.999, and 0.005, respectively. The algorithm stops when either the temperature drops to the terminal value or 5000 iterations are completed. SA was chosen for its robust ability to escape local optima and find a near-global minimum in a complex energy landscape, which is critical for ensuring high QL road extraction. As a result, the final fused road network is obtained by this optimal label configuration.

IV. Experimental Results and Discussion

A. Experimental Setup

1) Data Description: To evaluate the effectiveness and robustness of the proposed methodology, experiments were conducted on multitemporal and multimodal images. The primary characteristics of the test data utilized in the experiments are summarized in Table I. The multitemporal images consist of two optical data with a spatial resolution of 0.75 m, acquired by the Jilin-1 satellite over Wulan, Qinghai, China. The multimodal images are derived from the FUSAR-Map dataset [67]. The data were acquired over Jiujiang, Jiangxi, China, and include Gaofen-3 (GF-3) SAR imagery with HH polarizations, along with the corresponding optical image obtained from Google Earth.

TABLE I
Basic Information of Test Data

Study site	Imaging time	Sensor	Space resolution	Image size
Wulan	Nov. 2021 (S_{t1}) / Apr. 2022 (S_{t2})	Jilin-1	0.75m	7321×6560
Jiujiang	Aug. 2016	GF-3 (S_{sar}) / Google Earth (S_{opt})	1.00m	5632×7168

These data were selected to evaluate the proposed framework under diverse scenarios. As shown in Figs. 6 and 8, Case Studies I, and II respectively, cover urban areas characterized by dense buildings, heterogeneous road materials, and complex spectral signatures, which pose significant challenges for accurate road extraction. In addition, the test sites contain representative regions with occlusion caused by tree canopies, building shadows, and other urban structures, as highlighted in Fig. 7. These characteristics enable a comprehensive assessment of the proposed method in complex environments with significant occlusion.

2) Evaluation Metrics: To assess the performance of our method, we conduct both visual and numerical analyses of the results presented in this article. Six recently applied and representative decision fusion methods are used for comparison with the proposed method: logical OR fusion (OR) [10], naive Bayes (NBs) [13], SVM [15], Dempster–Shafer (DS) theory [22], Bayesian network fusion (BNF) [14], and DNNs [19]. For the visual evaluation, the road network extraction outcomes are qualitatively compared with the reference road network, focusing on factors such as road length, shape, junctions, and connectivity. The reference road network was created using ground truth data manually annotated from the original satellite imagery.

For the numerical comparison, the evaluation metrics of completeness (CP), CR, QL and RD are calculated using the reference data, following the methodology outlined in [68]. According to the approach in [68], a buffer is applied to determine the matching segments between two road networks. In this study, the buffer is set to five pixels. The specific definitions of the aforementioned evaluation metrics are provided by

$$CP = L_{MR}/(L_{MR} + L_{UR}) \tag{16}$$
$$CR = L_{ME}/(L_{ME} + L_{UE}) \tag{17}$$
$$QL = L_{ME}/(L_{ME} + L_{UE} + L_{UR}) \tag{18}$$
$$RD = |L_{ME} - L_{MR}|/L_{ME} \tag{19}$$

where L_{MR}, L_{UR}, L_{ME}, and L_{UE} are the length of matched reference, the length of unmatched reference, the length of matched extraction, and the length of unmatched extraction, respectively. The QL serves as a composite metric that

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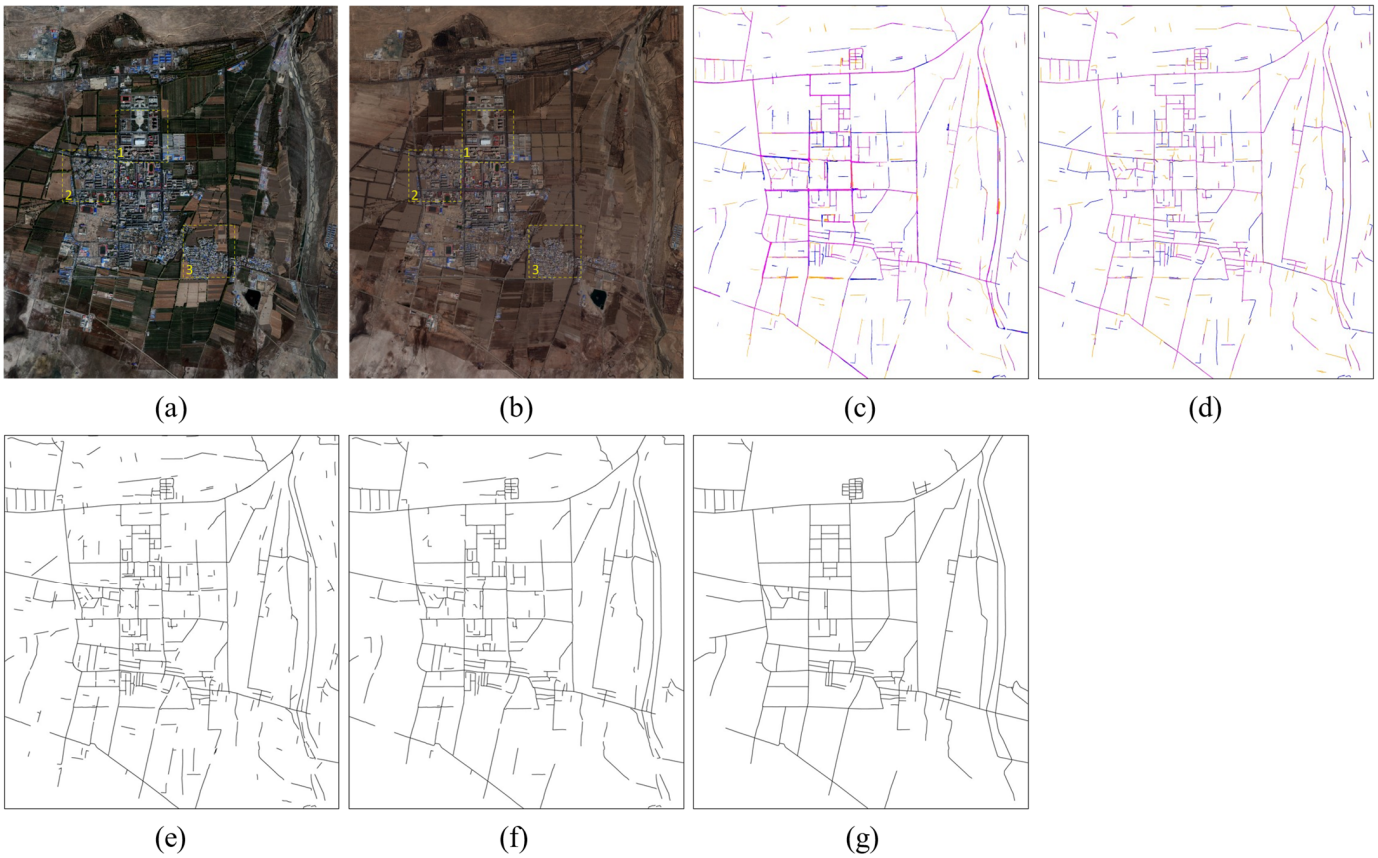


Fig. 6. Experimental results for Case Study I (the yellow dotted boxes are three subregions for detailed comparison). (a) Original optical image taken in Nov. 2021 (St1). (b) Original optical image taken in Apr. 2022 (S_{t2}). (c) Segmented road maps from S_{t1} and S_{t2} (purple: overlapping part; orange: areas unique to S_{t1}; blue: areas unique to S_{t2}). (d) Road thinning maps from S_{t1} and S_{t2}. (e) Result of the first-stage fusion. (f) Result of the second-stage fusion. (g) Ground truth data.

integrates both *CP* and *CR*. *RD* refers to the proportion of correctly matched extractions that are duplicated, meaning they overlap with themselves. It can be inferred that higher *QL* and lower *RD* values indicate that the extraction result is more consistent with the reference data. In addition, the total running time (RT) using different methods is also proposed in this article. It should be noted that the reported times only include the decision-level fusion stage; data preparation and training times are excluded from the comparison. All the experiments were conducted on a computer with an AMD R7-4800 H processor and 16 GB RAM using MATLAB implementations. The source code of the proposed method is publicly available at: https://github.com/Shaw-22/RoadFusion-ErTc

B. Case Study I: Multitemporal Images

1) Implementation Details: The first case study focuses on multitemporal optical images which were acquired during different seasons. The original test images are shown in Fig. 6(a) and (b). For data preparation of the proposed framework, we use the MSMDFF-Net model [58], trained on the DeepGlobe [69] dataset, to segment roads in each optical image separately. This yields two segmented road maps [see Fig. 6(c)] along with their corresponding confidence maps. Fig. 6(d) presents the results of further road thinning using the method described in [60]. Notably, Fig. 6(c) and (d) is the superimposed display of two extracted road maps, where purple indicates overlapping part, orange denotes areas unique to S_{t1}, and blue corresponds to areas unique to S_{t2}. In the following steps, we combine these multitemporal outputs to form one road network using our decision fusion framework.

For the parameter setting, the proposed method primarily depends on the BP functions, reliability r, and weight w in the ER rule, as well as the factor γ in (11) that defines the energy function. Since S_{t1} and S_{t2} both apply the same model to predict the road segmentation map for fusion, their BP functions are identical, as illustrated in Fig. 5, which is derived from (8). The reliabilities r_{t1} and r_{t2} are also the same, set to 0.78 based on the F1 score from the experimental results in [58]. Given that the images are acquired using the same sensor and within a close temporal window, the weights are assigned as $w_{t1} = w_{t2} = 1$. The factor γ in the energy function is empirically set to 0.3.

2) Experimental Results and Evaluation: According to Fig. 6(c) and (d), significant differences can be observed in the road segmentation results of multitemporal images when using the same model. On one hand, variations in vegetation growth across seasons lead to differences in image tone. On the other hand, the low sun angle during morning and evening hours can cast shadows from tall buildings or vegetation, potentially obstructing the road and affecting the segmentation results.

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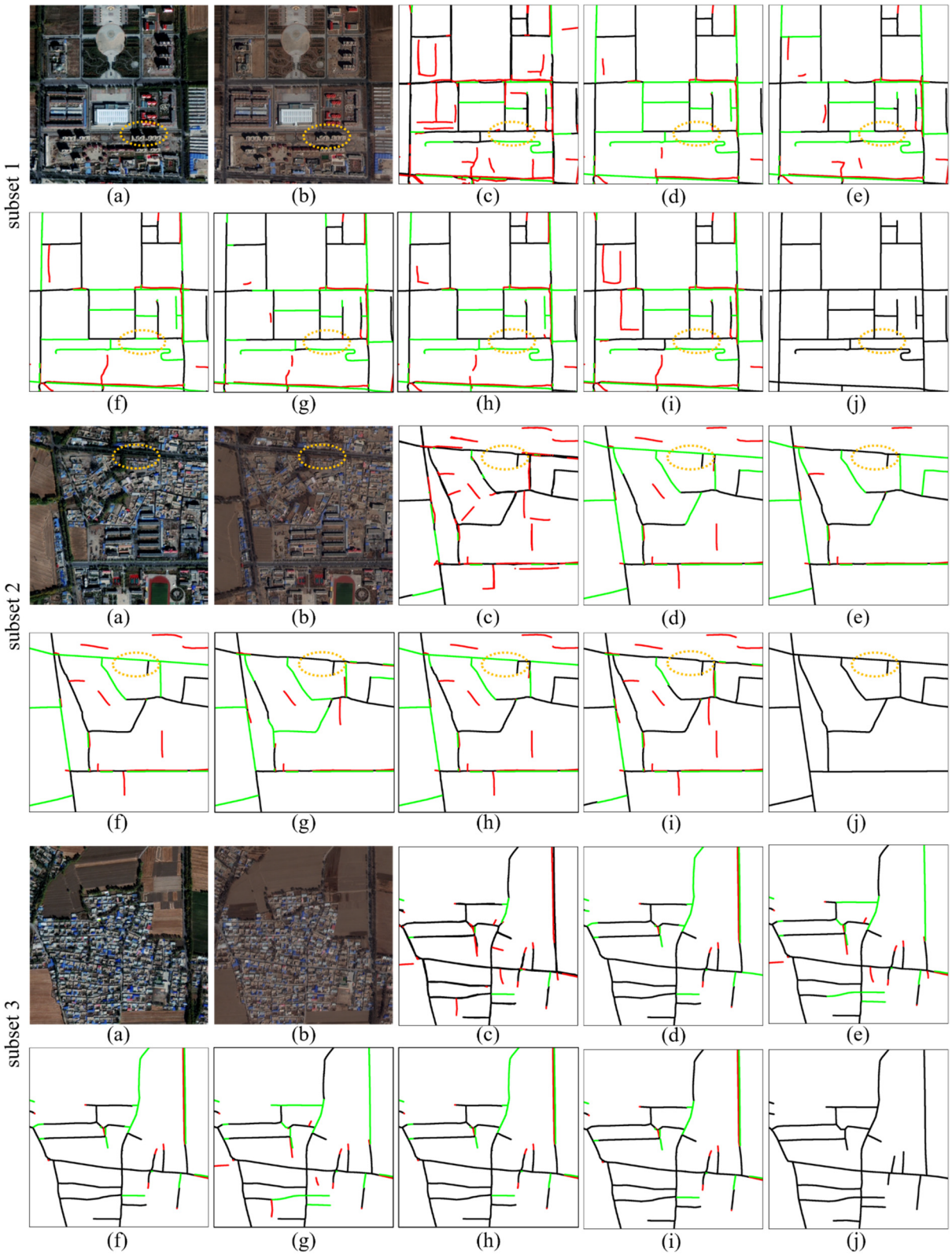


Fig. 7. Qualitative comparison of our method with several other established fusion methods on three subregions from Case Study I (black: correct extraction; red: incorrect extraction; green: missing extraction). (a) Original optical image taken in Nov. 2021 (S_{t1}). (b) Original optical image taken in Apr. 2022 (S_{t2}). (c) Result of OR method. (d) Result of NB method. (e) Result of SVM method. (f) Result of DS method. (g) Result of BNF method. (h) Result of DNN method. (i) Result of the proposed method. (j) Ground truth data.

Therefore, the use of multitemporal RS images for fusion is not only necessary but also scientifically justified to improve the accuracy of road extraction.

The result obtained after the proposed first-level fusion is presented in Fig. 6(e). The visual extraction results indicate that our method effectively removes overlapping and redundant

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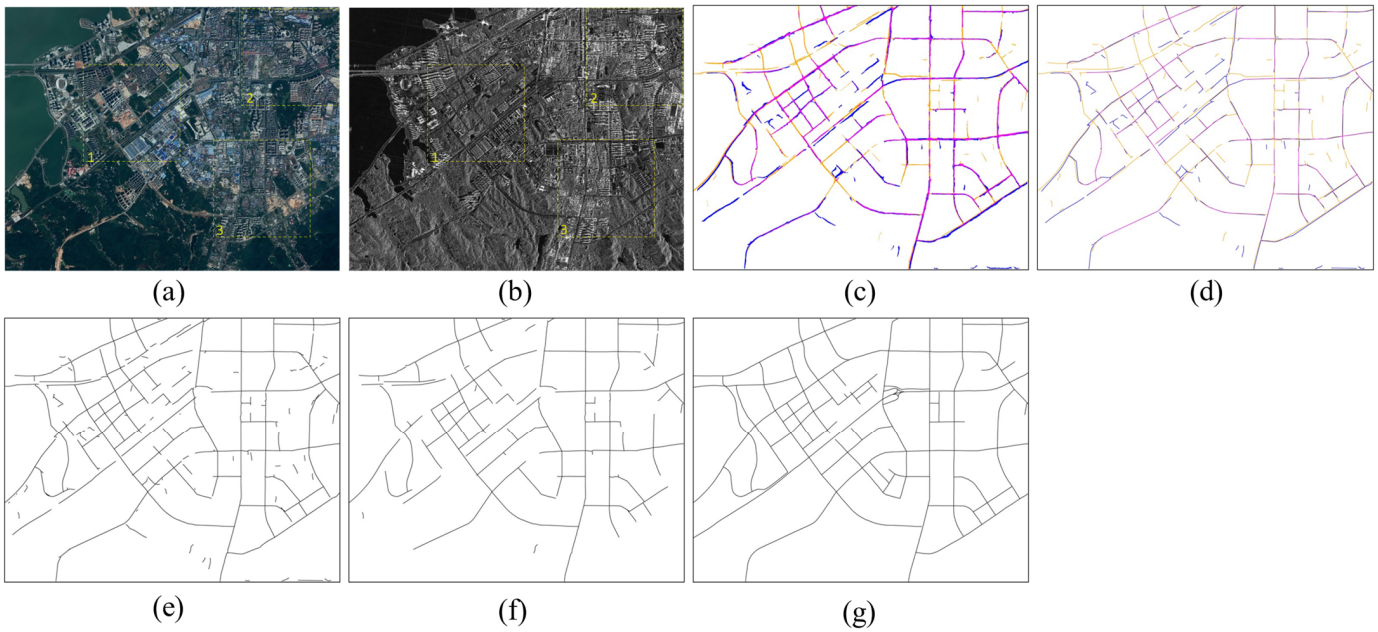


Fig. 8. Experimental results for Case Study II (the yellow dotted boxes are three subregions for detailed comparison). (a) Original optical image (S_{opt}). (b) Original SAR image (S_{sar}). (c) Segmented road maps from S_{opt} and S_{sar} (purple: overlapping part; orange: areas unique to Sopt; blue: areas unique to S_{sar}). (d) Road thinning maps from S_{opt} and S_{sar}. (e) Result of the first-stage fusion. (f) Result of the second-stage fusion. (g) Ground truth data.

TABLE II
COMPARISON OF ROAD NETWORK EXTRACTION FROM SINGLE-TEMPORAL VERSUS MULTITEMPORAL IMAGES

Source	*CP* (%)	*CR* (%)	*QL* (%)	*RD*
S_{t1}	68.68	69.35	52.69	0.0094
S_{t2}	74.91	70.09	56.77	0.0086
S_{t1} & S_{t2}	**77.58**	**73.35**	**60.52**	**0.0056**

TABLE III
COMPARISON OF OUR PROPOSED METHOD WITH SEVERAL OTHER ESTABLISHED METHODS ON THE TEST MULTITEMPORAL IMAGES

Methods	*CP* (%)	*CR* (%)	*QL* (%)	*RD*	*RT*(s)
OR	**85.67**	68.85	**61.74**	0.3543	0.18
NB	61.95	**81.55**	54.33	0.0109	91.27
SVM	67.32	74.12	54.51	0.0106	90.69
DS	68.63	76.61	56.74	0.0073	128.54
BNF	68.35	77.10	56.82	0.0084	96.32
DNN	66.88	76.22	55.33	0.0110	88.69
Proposed	77.58	73.35	60.52	**0.0056**	387.57

road segments, while retaining comprehensive road information extracted from multiple sources. Fig. 6(f) shows the visual results of the proposed second-level fusion on the test images. Compared with the ground truth in Fig. 6(g), it is evident that the incorporation of multisource fusion and TCs leads to a notable reduction of incorrect road segments.

Table II presents the quantitative evaluation results of road network extraction based on single-temporal and multiple-temporal imagery. It can be observed that the *CR* values for road extraction using single-temporal images are relatively similar, whereas the *CP* value for S_{t2} is significantly higher than that for S_{t1}. As expected, the fusion of road extraction results from multitemporal images yields the best performance across all the four metrics. This confirms that integrating multiple complementary information sources enhances the accuracy of road extraction.

3) Comparison With the Existing Methods: The quantitative experimental results of our proposed method compared to several other established approaches on Case Study I are shown in Table III. Regarding the *CP* and *QL* metrics, the OR method achieves the highest scores among all compared methods. This can be attributed to its simple union-based fusion strategy, which maximizes the preservation of candidate detections and leads to very high *CP*, thereby resulting in an elevated *QL* value. However, this advantage comes at the cost of substantial RD, as reflected by the highest *RD* value. Due to the lack of mechanisms for removing duplicate structures or enforcing geometric consistency, the OR method often produces overlapping segments and noisy artifacts, limiting its practical applicability. The NB method has the highest *CR* of 81.55%, albeit with a lower CP (61.95%), indicating high CR but with numerous missed detections. While the SVM method attains a comparable *QL* to NB, this comes at the cost of reduced *CR* and enhanced *CP*. The DS method exhibits slight improvements in both *CP* and *CR* over SVM. Additionally, the BNF and DNN methods demonstrate similar performance trends to DS, with QL values around 56.82% and 55.33%, respectively. While these methods slightly enhance the CP compared to SVM, they still struggle to effectively suppress noise or accurately recover connectivity in complex scenarios. In contrast, the proposed method achieves the highest *QL* value among all low RD methods, while maintaining a balanced and relatively high performance in both *CP* and *CR*.

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In terms of running time (RT), the OR method remains the most efficient due to its simple implementation, which involves only a direct set union without iterative optimization or model inference. By comparison, the proposed method requires a longer running time, mainly due to the global optimization process described in (11). Nevertheless, this additional computational cost is justified by the improved structural QL and reduced RD of the results. To facilitate large-scale or real-time applications, the proposed framework could be further enhanced by incorporating more efficient optimization paradigms. Potential strategies include parallelized patch-based processing or the adoption of computationally efficient solvers, such as graph cuts, to strike a better balance between fusion QL and processing speed.

Fig. 7 shows the qualitative comparison between our proposed method and other approaches, using three representative subset regions selected from the original images. The results of OR show a high density of line segments, suggesting considerable overlap between the extracted road segments. Among the other methods, the NB method achieves the best performance in terms of incorrect detection (highlighted in red). However, when focusing solely on missing segments (highlighted in green), the proposed method has the fewest omissions. As highlighted by the orange dashed ellipses in Fig. 7, single-source road extraction methods often fail in regions affected by building shadows or vegetation occlusion, resulting in fragmented or missing road segments. In contrast, the proposed method leverages complementary information from multisource data, effectively alleviating these information blind spots and enabling more complete road extraction in such challenging areas. Overall, our results align most closely with the ground truth and yield superior road network connectivity due to the incorporation of TCs.

C. Case Study II: Multimodal Images

1) Implementation Details: For the experiments on the multimodal data, satellite images acquired from both optical and SAR sensors over the same geographic region are utilized. As illustrated in Fig. 8(a) and (b), the SAR and optical images differ notably. To segment road targets, the D-LinkNet34 models [57] are trained separately on the SAR and optical datasets. A comparative overlay of the segmentation results is presented in Fig. 8(c), while Fig. 8(d) shows the corresponding comparison of the road thinning results.

Fig. 9 shows the BP functions of the ER rule used in the proposed decision-level fusion stage, which we derive from the sample data using (8). Based on the F1 scores obtained on the test set using the well-trained D-LinkNet34 models, the reliability parameters r_{opt} and r_{sar} are determined to be 0.72 and 0.52, respectively. Given the higher spatial resolution of the optical images and the more prominent road features they exhibit, a higher fusion weight is assigned to the optical modality. Specifically, the weights are set as $w_{opt} = 1$ for the optical data and $w_{sar} = 0.7$ for the SAR data. Following the configuration used in Case Study I, the parameter γ of the energy function is set to 0.3 in this experiment.

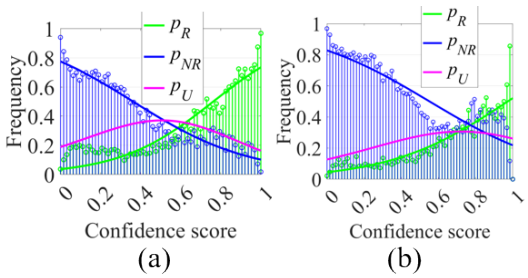


Fig. 9. BP functions used in Case Study II. (a) BP functions for optical image. (b) BP functions for SAR image.

TABLE IV
COMPARISON OF ROAD NETWORK EXTRACTION FROM SINGLE-MODAL VERSUS MULTIMODAL IMAGES

Source	*CP* (%)	*CR* (%)	*QL* (%)	*RD*
S_{opt}	71.00	73.43	56.49	0.0031
S_{sar}	44.02	54.39	32.15	0.0106
S_{opt} & S_{sar}	**72.83**	**88.16**	**66.34**	**0.0036**

2) Experimental Results and Evaluation: Fig. 8 presents the road maps before the decision-level fusion, the fused road maps generated using the proposed framework, and the corresponding reference road map. Among them, Fig. 8(e) shows the result after the first-level fusion, which significantly reduces redundant road segments. Fig. 8(f) illustrates the outcome of the second-level fusion, which effectively integrates evidential information from multimodal data along with topological prior knowledge of the road network.

Moreover, we compare road extraction results from single-modal and multimodal images. As evidenced by the quantitative results in Table IV, optical imagery outperforms SAR across all evaluated road extraction metrics. Specifically, in terms of the *QL* index, the optical is about 24% higher than the SAR, suggesting that optical images are generally more suitable for road extraction tasks. Nevertheless, our decision-level fusion method surpasses using optical imagery alone: *CP* improves by approximately 2%, *CR* increases by around 15%, and RD remains low. These findings not only underscore the complementary nature of multimodal data but also validate the effectiveness and robustness of the proposed fusion framework.

3) Comparison With the Existing Methods: We tested five strategies toward road fusion using multimodal images and evaluated their performance. Fig. 10 provides a qualitative comparison of the results. Among all the evaluated methods, the logical OR combination produces significantly more redundant road segments. The NB and SVM fusion methods result in a higher number of falsely extracted roads (highlighted in red). Regarding missing road segments (highlighted in green), the DS approach exhibits the highest omission rate. In terms of the total number of false positives and missing detections, the proposed method demonstrates clear advantages, indicating its superior performance and reliability in road fusion tasks.

The quantitative evaluation presented in Table V further supports the aforementioned conclusions. It is evident that the logical OR method achieves the highest *CP*, but at the cost of

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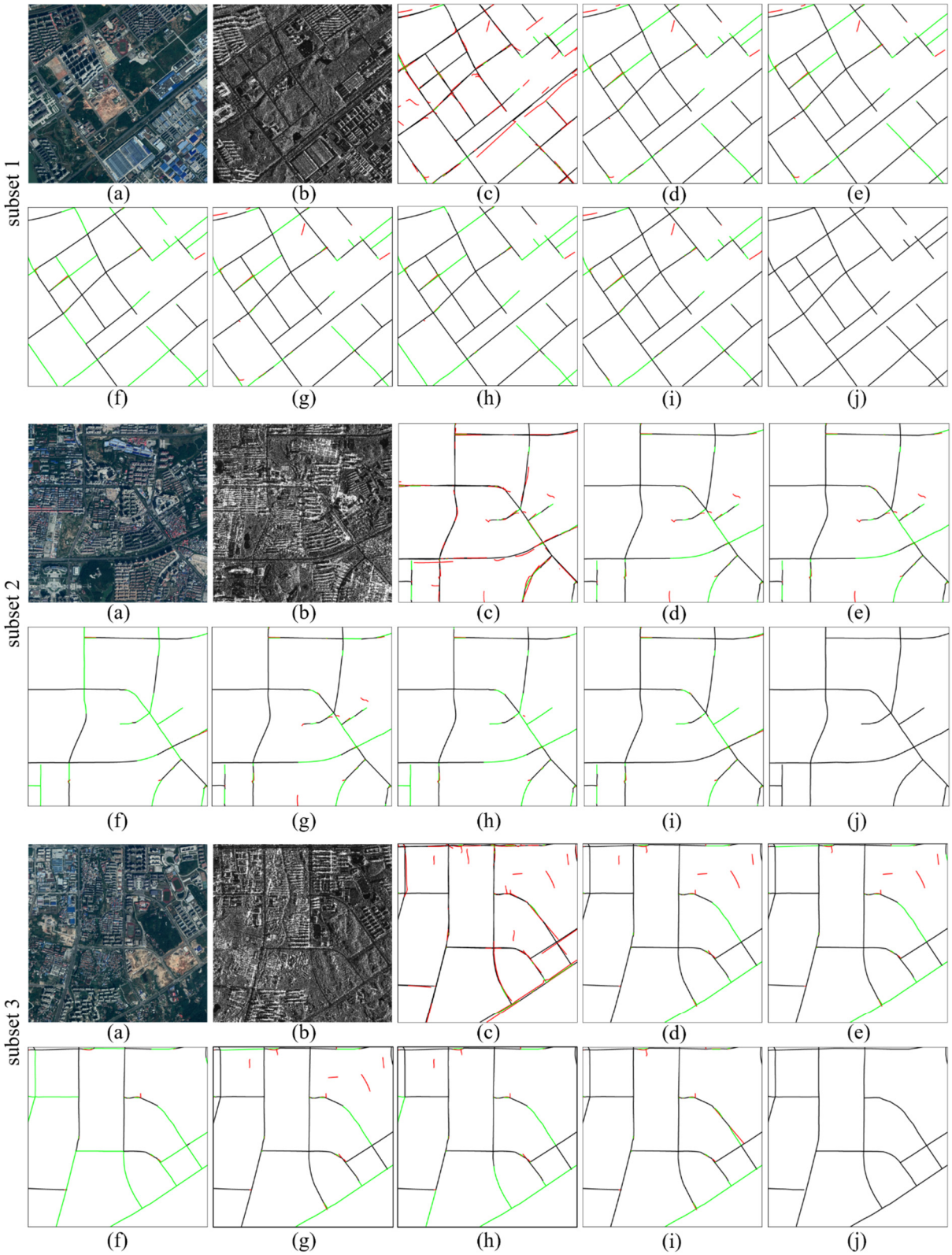


Fig. 10. Qualitative comparison of our method with several other established fusion methods on three subregions from Case Study II (black: correct extraction; red: incorrect extraction; green: missing extraction). (a) Original optical image (S_{opt}). (b) Original SAR image (S_{sar}). (c) Result of OR method. (d) Result of NB method. (e) Result of SVM method. (f) Result of DS method. (g) Result of BNF method. (h) Result of DNN method. (i) Result of the proposed method. (j) Ground truth data.

the lowest *CR* and the highest RD, consistent with the results observed in Case Study I. The NB and SVM methods exhibit similar performance, with NB performing slightly better across all evaluation metrics. Notably, in this experiment, the DS

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TABLE V

COMPARISON OF OUR PROPOSED METHOD WITH SEVERAL OTHER ESTABLISHED METHODS ON THE TEST MULTIMODAL IMAGES

Methods	*CP* (%)	*CR* (%)	*QL* (%)	*RD*	*RT*(s)
OR	**82.58**	64.13	56.49	0.2548	0.15
NB	71.52	86.15	64.14	0.0046	49.80
SVM	69.38	85.97	62.32	0.0051	44.99
DS	44.55	**93.55**	43.22	0.0091	64.22
BNF	58.08	93.07	55.67	0.0067	46.23
DNN	70.09	86.49	63.17	0.0051	43.05
Proposed	72.83	88.16	**66.34**	**0.0036**	184.90

TABLE VI

COMPONENTWISE ABLATION STUDY ON THE PROPOSED DECISION-LEVEL FUSION FRAMEWORK FOR ROAD NETWORK EXTRACTION ACROSS TWO CASE STUDIES

Case study	Methods			*CP* (%)	*CR* (%)	*QL* (%)	*RD*
	GR	ER	TC				
I	√			**79.07**	63.83	54.60	**0.0046**
	√	√		68.20	**76.49**	56.39	0.0074
	√		√	77.47	71.42	59.14	0.0046
	√	√	√	77.58	73.35	**60.52**	0.0056
II	√			78.01	73.85	61.13	**0.0009**
	√	√		65.70	**91.21**	61.79	0.0051
	√		√	**78.09**	78.63	64.42	0.0028
	√	√	√	72.83	88.16	**66.34**	0.0036

method yields the lowest *CP* but the highest *CR* among all compared methods. This may be attributed to the DS method's equal weighting of optical and SAR data during the fusion process. The BNF method achieves a high *CR* comparable to DS, while maintaining a moderate *CP* and relatively low RD, indicating its effectiveness in suppressing false positives at the expense of CP. The DNN-based method shows a more balanced tradeoff between *CP* and *CR* and achieves competitive *QL* performance; however, its improvement remains limited by the insufficient exploitation of complementary multisource information.

In summary, although the proposed method does not attain the highest values in either *CP* or *CR* (ranking second in both), it achieves the best performance in terms of the overall QL index (*QL*), surpassing the second-best method by approximately 2%. This demonstrates the superior balance and effectiveness of the proposed fusion approach. The running time is longer (184.90 s), primarily due to the global optimization process; nevertheless, the gain in extraction QL indicates promising applicability in practical scenarios where accuracy is prioritized.

D. Componentwise Performance Evaluation

This article proposes a modular and generic decision-level fusion framework for road extraction from multisource RS images. The framework consists of two stages: the first stage involves fusion based on geometric rules (GRs), while the second stage is the fusion integrating multisource ER rules and TCs. To evaluate the contribution of each component, we conducted ablation experiments, and the results from two case studies are summarized in Table VI. It is worth noting that the results corresponding to GR + ER and GR + TC can be conveniently derived by setting the parameter γ to 1 and 0 in (11), respectively.

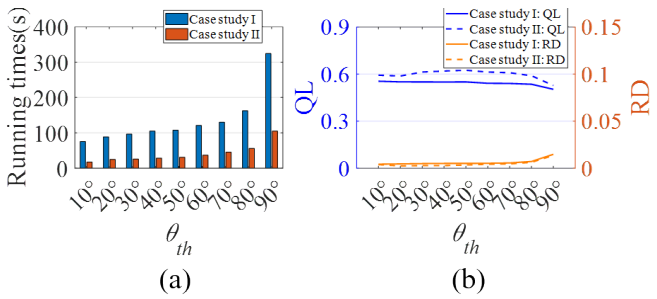


Fig. 11. Quantitative metrics with respect to the threshold θ_{th} in the experiments. (a) Variation in running time. (b) Variation in *QL* and *RD*.

Comparing the experimental data of the two case studies in Table VI, it can be seen that the *RD* index remains consistently low across all combinations of components. However, significant variations are observed in the accuracy-related metrics. In Case Study I, the baseline method employing only GR achieves a *CP* of 79.07% and a *CR* of 63.83%. When ER is introduced (GR + ER), *CR* increases substantially to 76.49%, although *CP* drops to 68.20%. This indicates that multisource evidence helps filter out false positives at the cost of reduced CP. In contrast, incorporating TC (GR + TC) also improves *CR* while maintaining a relatively high CP (77.47%), suggesting that topological information contributes to road connectivity with minimal sacrifice of detection capability. The full framework (GR + ER + TC) achieves the best overall QL (60.52%), confirming the complementary benefits of integrating all three components. Similar patterns are observed in Case Study II, reinforcing the framework's generalizability across different contexts. In summary, our ablation analysis confirms that each component contributes distinct value to the overall framework, and their joint integration substantially enhances the accuracy and robustness of road extraction from multisource RS imagery.

E. Parameter Analysis of the First-Stage Fusion

Line-NMS, introduced in Algorithm 1, plays a key role in the first-stage fusion process, with the threshold θ_{th} as its sole hyperparameter. To explore the impact of θ_{th} on road fusion performance, we vary its value and evaluate the corresponding first-level fusion results on both Case Study I and Case Study II.

Fig. 11 illustrates the influence of the parameter θ_{th} on three key metrics—running time, *QL*, and *RD*—across the two case studies. As θ_{th} increases from 10° to 90°, the running time exhibits a growing trend, particularly in Case Study I, where a sharp increase is observed beyond 70°. In contrast, *QL* remains relatively stable for both cases, and *RD* values remain consistently low, with only minor fluctuations. Overall, the results indicate that θ_{th} has minimal impact on extraction QL and RD but substantially affects computational cost. Therefore, a smaller θ_{th} is generally recommended for faster processing.

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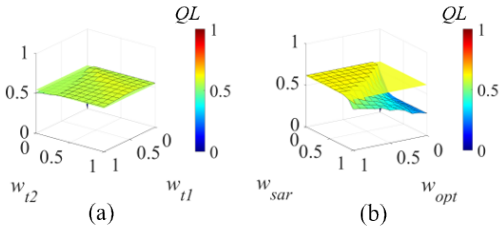


Fig. 12. Quantitative metrics with respect to weight w in ER (the nongridded plane acts as a reference, indicating the QL index resulting from the first-stage fusion). (a) Experimental result in Case Study I. (b) Experimental result in Case Study II.

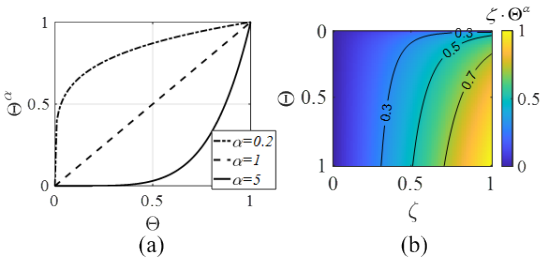


Fig. 13. Analysis of parameter α in TC. (a) Effect of parameter α on the energy term Θ^α with respect to angle attribute Θ. (b) Distribution of $u_p = \zeta \cdot \Theta^\alpha$ under different ζ and Θ with $\alpha = 0.2$.

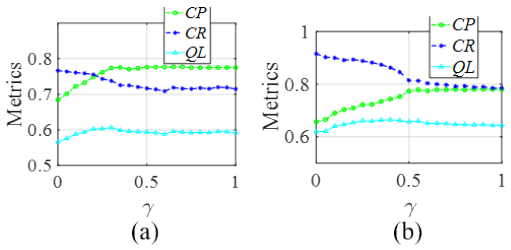


Fig. 14. Quantitative indexes (CP, CR, and QL) versus γ in the energy function. (a) Results for Case Study I. (b) Results for Case Study II.

However, in scenarios where overlapping road segments form larger angles, a higher θ_{th} may be chosen to ensure effective fusion.

F. Parameter Analysis of the Second-Stage Fusion

1) Effect of Different Configurations for Weight w in ER: In the ER module of the proposed fusion framework, the weight w ($0 \le w \le 1$) represents the relative importance assigned to each data source. This parameter provides the flexibility to incorporate domain expertise or task-specific preferences, such as prioritizing high-resolution sensors over lower-resolution ones. Fig. 12 presents a sensitivity analysis of the ER-based fusion results under different weight configurations. For the multitemporal road fusion experiment in Case Study I, the two data sources exhibit comparable reliability. As shown in Fig. 12(a), the QL performance of ER-based fusion remains consistently superior to the first-level fusion results across a wide range of weight settings. Even under less favorable configurations, the performance degradation is marginal, indicating stable fusion behavior.

In contrast, Case Study II involves two data sources with markedly different characteristics [Fig. 12(b)]. Here, assigning a higher weight to the more informative source improves fusion performance. This observation confirms that w serves as a meaningful importance indicator that complements source reliability, rather than a tunable parameter optimized solely for performance gains.

2) Effect of Different Configurations for Parameters in TC: As presented in (14) and (15) in Section III-E, the TC component of the energy function contains four positive parameters: P_b, P_i, P_s, and α. First, for isolated segments, the energy change Δu_p when labeled as road (R) and NR can be calculated by

$$\begin{aligned}\Delta u_p &= u_p(l_i = R) - u_p(l_i = NR) \\ &= -\zeta_i + P_i. \end{aligned} \tag{20}$$

If $P_i > 1$, then Δu_p is always positive, leading to the removal of all isolated segments. Conversely, if P_i lies between 0 and 1, isolated segments with normalized lengths smaller than P_i will be removed. Similarly, analyzing Δu_p reveals that a value of $P_b > 1$ results in the elimination of all burrs. When P_b is within the range $(0, 1]$, shorter burrs or those forming small angles with the trunk road are more likely to be removed.

The parameter α controls the penalty associated with the angle between adjacent segments. As illustrated in Fig. 13(a), a larger value of α results in a higher u_p for segments with large Θ, indicating a stronger suppression effect. Given that real road networks typically exhibit low curvature, segments with small angle attribute ($\Theta < 0.25$) should be strongly suppressed, while those with large angle attribute should be preserved. Based on this observation, α is fixed at 0.2 throughout this article. Fig. 13(b) presents the distribution of $\zeta \cdot \Theta^\alpha$ under various values of ζ and Θ when $\alpha = 0.2$. The last parameter, P_s, serves to suppress intersections with more than four connected segments. It only needs to be a positive value and can be set to the same order as the other parameters.

3) Effect of Different Weights γ for ER and TC: As described in (12), the parameter γ ($0 \le \gamma \le 1$) controls the contribution of the topological constraint term in the energy function used during the second-level fusion process. Fig. 14 presents the fusion performance metrics for Case Studies I and II under varying values of γ. As γ increases from 0, the CP initially increases and then stabilizes, while the CR shows a decreasing trend before leveling off. The inflection points for both metrics appear around $\gamma = 0.5$. As for the overall performance (QL), it first increases with rising γ, then slightly decreases and eventually stabilizes. These results indicate that incorporating topological constraints can effectively enhance the overall performance of road network extraction. Moreover, the experimental results suggest that the optimal value of γ is approximately 0.3.

V. Conclusion

This article presents a novel decision-level fusion framework for road network extraction from multisource RS data. The proposed framework adopts a two-stage fusion strategy, consisting of a GR-based fusion stage and a topology-aware

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evidence fusion stage. In the first stage, a Line-NMS scheme is introduced to effectively eliminate overlap and RD among candidate road segments, thereby reducing the computational burden for subsequent fusion. In the second stage, instead of directly using soft classification outputs as belief assignments, we employ a data-driven sigmoid-based BP fitting strategy to mitigate bias in the initial evidence. We also explicitly embed road TCs into a graph-based optimization formulation, enforcing topological CR during multisource fusion rather than as a separate postprocessing step. As a result, the proposed framework produces a more coherent and reliable road network and demonstrates robust performance without requiring coregistered multisource training data.

Experimental evaluations on multitemporal and multimodal RS imagery demonstrate that the proposed method significantly improves the QL of road network extraction, outperforming both individual-source results and several existing decision-level fusion approaches. Moreover, the framework proves readily applicable to practical scenarios, such as automatic updating of large-scale road network databases, even when road information originates from different sensors or ML model outputs. Future work will focus on accelerating computational efficiency and extending the framework to other types of infrastructure extraction tasks, such as railways and buildings.

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